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Assignment #5 *The answers should be submitted until Wednesday November 11, 14:00*

Exercise 1

Pnstab02den

Consider the polynomial $p(x) = (x-2)^5$. In expanded form, it can be written as

$$f(x) = x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32.$$

1. Define two Python functions implementing these expressions directly. Using NumPy floating-point arrays, evaluate $p(x)$ and $f(x)$ for x from 1.998 to 2.002, with an increment of 0.0001. You may use `np.linspace(1.998, 2.002, 41)` to include both endpoints.
2. Represent $f(x)$ and $p(x)$ graphically in the same figure using Matplotlib. Distinguish the two expressions with different colours or markers, and include a legend. What do you observe?
3. Repeat point 1 with x from 1000.998 to 1001.002, keeping an increment of 0.0001. In another figure, plot the difference $f(x) - p(x)$ against x. Comment on the results.

Exercise 2

Pnstab03ben

Consider the numerical sequence

$$S_n = u_1 + u_2 + \cdots + u_n, \qquad u_i = \left(\frac{1}{4}\right)^i.$$

In exact arithmetic, the following relationship holds for every $n \geq 1$:

$$R_n = \frac{1 - 3S_n}{u_n} = 1.$$

Using Python, compute u_n, S_n and R_n for $n = 1, \ldots, 30$ from the recurrence

$$\begin{aligned} u_n &= \frac{1}{4} u_{n-1}, \\ S_n &= S_{n-1} + u_n, \\ R_n &= \frac{1 - 3S_n}{u_n}, \end{aligned}$$

with $u_0 = 1$ and $S_0 = 0$.

Store the values in NumPy floating-point arrays. You may allocate arrays of length 31 so that index n corresponds directly to the mathematical subscript n, including the initial values at index 0.

Plot R_n, S_n and u_n against $n = 1, \ldots, 30$ in the same figure, using a legend to identify the three sequences. Comment on the results.

Exercise 3

Pnstab01den

Create a NumPy floating-point array y containing $n = 8000$ observations defined by

$$y_i = 2z_i + 10^9,$$

where the z_i are independent standard normal random variables. For reproducibility, initialise the random generator once with `rng = np.random.default_rng(1)` and generate the standard normal observations using `rng.standard_normal(n)`. Use the same observations throughout the exercise.

1. Compute an estimate of the variance of y using the following formula (the raw-moment one-pass algorithm):

$$\widehat{\sigma}_y^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} y_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} y_i\right)^2\right].$$

   Implement this expression directly, rather than using `np.var`. Knowing that the theoretical variance is 4, comment on the result. Compute the relative deviation from this value:

$$e_{\text{rel}} = \frac{|\widehat{\sigma}_y^2 - 4|}{4}.$$

2. In exact arithmetic, the transformation $x_i = y_i - d$, where d is constant, leaves the variance unchanged. Estimate the variance of x using the formula from point 1 for different values of d, for example $d = 10^k$ with $k = 0, 1, \ldots, 10$. Compute the relative deviation from 4, comment on the results, and give a criterion for choosing d.

3. An alternative method first computes the mean

$$\overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i,$$

   and then evaluates

$$\widehat{\sigma}_y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})^2.$$

   This method is called the two-pass algorithm. Implement it directly and compute the relative deviation from 4.

4. Why are these algorithms called one-pass and two-pass algorithms? Comment on the advantages and disadvantages of each method.

Interpretation: A finite random sample need not have variance exactly 4, even in exact arithmetic. The deviation from 4 therefore reflects sampling variability as well as numerical error. Compare the algorithms on the same sample to discuss their numerical behaviour.